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Innovation

“XR And Smart Glasses Are At An Inflection Point In India”

What is Kaynes Technology, a leading publicly listed electronics-manufacturing company, doing in the extended reality (XR) segment? EFY's Vidushi spoke with Himanshu Verma from Kaynes to find out.

Can you explain what the AR waveguide manufacturing line contains and why it is important for India?

Waveguides are essentially the screens for smart glasses and heads-up displays used in automotive applications. A display has two main parts: a projector and a waveguide. The projector sends light through the waveguide so the image appears on the glass. In simple terms, the waveguide acts as the display for smart glasses. It is a core component that relies on complex physics and advanced processes such as thin-film deposition, which are critical in both semiconductor and display manufacturing.

What makes DigiLens CrystalClear unique compared to other AR display technologies?

The key advantage of the proprietary material is that it delivers two major benefits. First, the luminous efficiency of the waveguides is extremely high, ensuring sharp and clearly visible displays even in very bright environments. The CrystalClear technology provides consistent brightness and clarity.

Second, it minimises eye glow, which can otherwise be socially noticeable when using smart glasses. This makes the experience more comfortable and natural for users, and the technology ranks among the best in reducing eye glow.

Kaynes also collaborates with DigiLens on manufacturing. The process enables high-yield, lower-cost production, addressing one of the industry's most common challenges.

Did you face any challenges in setting up India's waveguide line?

Doing something for the first time requires patience and specialised

expertise that is not easily available in the market. The right engineers and teams are needed to install and operate complex, high-value equipment. These were the primary challenges. While India has strong electronics capabilities, optics is an area where further competency still needs to be developed.

How is critical optics knowledge being shared and implemented in India?

DigiLens has a strong IP portfolio, including hundreds of patents in volumetric holographic grating technologies that underpin waveguide core technologies. Knowledge behind these patents is being shared with Kaynes so the company can build in-house capability, particularly in manufacturing processes. All key machines from DigiLens are now installed at the Mysuru and Chamara Nagar plants, and operators have developed full expertise in running them.

How did the partnership develop, and what synergies did DigiLens bring?

There is strong alignment between Kaynes and DigiLens. DigiLens focuses primarily on IP and technology development with deep domain expertise.

Kaynes contributes to manufacturing excellence. While DigiLens has developed advanced technologies and products, it requires support to scale and optimise manufacturing. By partnering with Kaynes, DigiLens benefits from proven manufacturing capabilities to efficiently and at scale produce and refine waveguides.

Do you partner with universities, provide training for engineers, or hire directly?

Much of the workforce comes

from local technical universities. Many fresh graduates are hired and receive extensive training to develop the skills required for advanced manufacturing.

For waveguides specifically, entry-level engineers work alongside PhD-level experts in optics and lasers because these systems are highly complex. The aim is not only to build a skilled workforce but also to ensure the right expertise is available to operate the systems and train the next generation for emerging technologies.

Which industry shows the highest demand for XR and immersive display technologies?

The automotive sector currently shows the highest demand. Most client requirements for these technologies come from automotive applications.

What message would you like to convey to investors, consumers, or tech enthusiasts about India's role in the XR evolution?

I would say XR and smart glasses are at an inflection point in India, similar to where mobile phones were 30 to 40 years ago, when screens were expanding, and touch interfaces were emerging. If this opportunity is captured, smart glasses could define the next three to four decades and potentially succeed smartphones.

India has the opportunity to lead, but doing so will require collaboration to develop more XR products, technologies, and applications. Based on interactions across China, the UK, and the USA, India is advancing rapidly. Kaynes is positioned to support and accelerate this growth through strong manufacturing capability and strategic partnerships. **EFY**